



Lab 05: Routing Between Different Networks

Router0 (2911) · Switch0 · Switch1 · PC0 · Laptop0 · PC1

Lab #05

Topic: Inter-Network Routing

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

1. Lab Objectives

- Connect two switches to a single 2911 router on different Gigabit interfaces.
- Place PC0 and Laptop0 on Switch0's network, PC1 on Switch1's network.
- Configure static routes or use connected routes for inter-network communication.
- Verify that devices on different subnets can reach each other through Router0.
- Relate to Network+ objective 1.4: Routing concepts and path determination.

2. Network Topology

Router0 (2911) sits at the top with two Gigabit interfaces. Gig0/0 connects to Switch0 (2960-24TT) which has PC0 (Fa0/1) and Laptop0 (Fa0/3). Gig0/1 connects to Switch1 (2960-24TT) which has PC1 (Fa0/1). Three different subnets are in use.

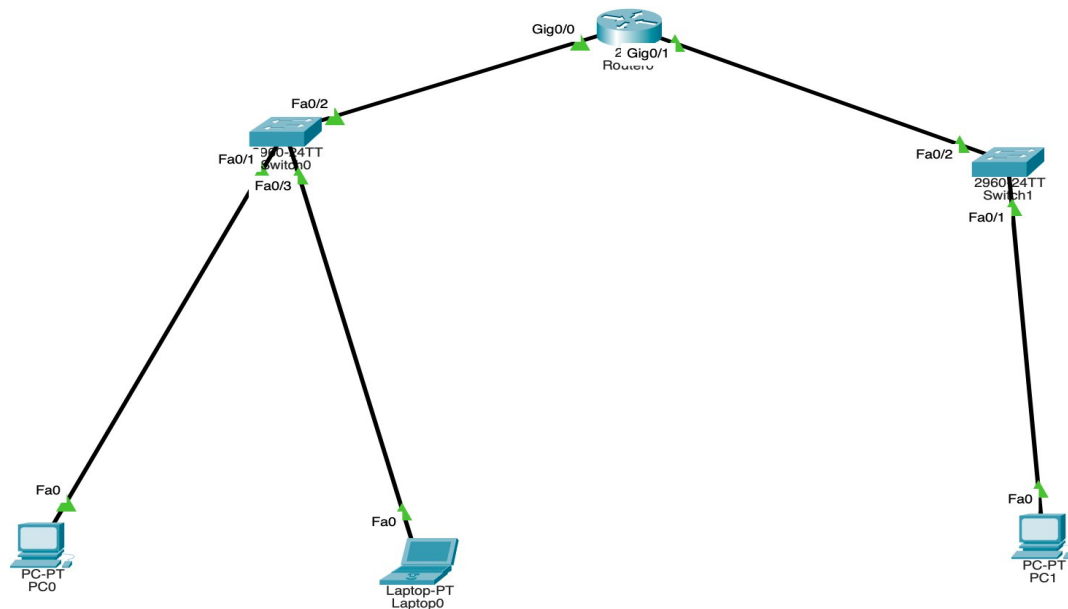


Figure 1 – Router0 connecting two switches: Switch0 (PC0+Laptop0) and Switch1 (PC1)

Device	Interface	IP Address	Subnet	Gateway
Router0	Gig0/0	192.168.1.1	192.168.1.0/24	—
Router0	Gig0/1	192.168.2.1	192.168.2.0/24	—



PC0	Fa0	192.168.1.10	192.168.1.0/24	192.168.1.1
Laptop0	Fa0	192.168.1.20	192.168.1.0/24	192.168.1.1
PC1	Fa0	192.168.2.10	192.168.2.0/24	192.168.2.1

3. Step-by-Step Configuration

3.1 Router0

```
Router0(config)# interface GigabitEthernet0/0
Router0(config-if)# ip address 192.168.1.1 255.255.255.0
Router0(config-if)# no shutdown
Router0(config-if)# exit
Router0(config)# interface GigabitEthernet0/1
Router0(config-if)# ip address 192.168.2.1 255.255.255.0
Router0(config-if)# no shutdown
Router0(config-if)# exit
Router0# write memory
```

4. Verification

```
PC0> ping 192.168.1.20      ! PC0 to Laptop0 (same subnet)
PC0> ping 192.168.2.10      ! PC0 to PC1 (cross-subnet via Router0)
Laptop0> ping 192.168.2.10 ! Laptop to PC1
Router0# show ip route      ! C = connected routes for both subnets
```

5. Key Takeaways

- One router with two interfaces creates two separate broadcast domains / subnets.
- PC0 and Laptop0 share the same subnet (192.168.1.0/24) — the switch handles their traffic.
- PC1 is on a different subnet — only the router can forward packets between them.
- Connected routes are added automatically when interfaces are configured and active.
- A laptop in Packet Tracer can use either a wired or wireless NIC — in this lab it is wired.